STAT 335, **Unit 3**: Time Series Regression and Exploratory Data Analysis

(Part b): Ordinary Least Squares Regression for Time Series

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Regression for Time Series

We want to look at the situation where a time series, say, x_t, for $t = 1, \ldots, n$, is possibly being influenced by a collection of other time series, say

$$z_{t1}, z_{t2}, \ldots, z_{tq}.$$

These independent time series are called **exogenous variables** (since they are "external" to the series x_t).

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These independent time series are called **exogenous variables** (since they are "external" to the series x_t).

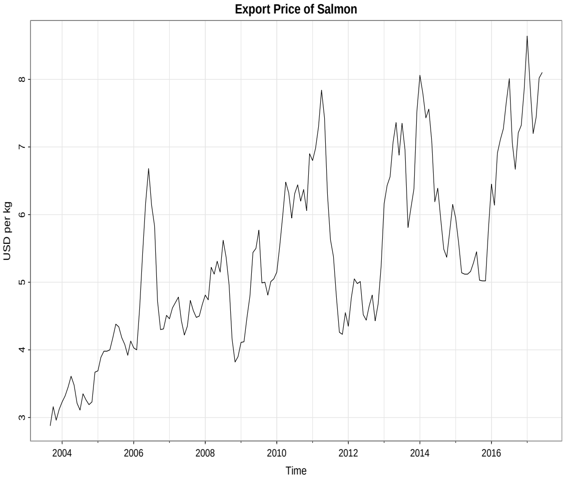
So, we're looking at a *linear regression model* of:

$$x_t = \beta_0 + \beta_1 z_{t1} + \beta_2 z_{t2} + \cdots + \beta_q z_{tq} + w_t,$$

where the $\beta_0, \beta_1, \ldots, \beta_q$ are unknown fixed regression coefficients and w_t is a white noise time series with variance σ_w^2.

Example: Estimating the Linear Trend of a Commodity

In the **astsa** package is a dataset **salmon** that contains the monthly export price of farm bred Norweigian salmon, from September 2003 to June 2017, in US dollars per kilogram.



Example: Estimating the Linear Trend of a Commodity

We want to estimate the *linear trend* of this time series by doing a linear regression on this data and finding the best fit line.

I.e., we'll fit the model:

$$x_t = \beta_0 + \beta_1 z_t + w_t,$$

where:

- w_t is the error term, which we will assume is modeled by white noise (may not be accurate but we will assume for now)
- z_t will be time, measured in months, since the x_t series is given in monthly increments

Example: Estimating the Linear Trend of a Commodity

Let's look at the **salmon** time series data set

Exercise 3b.1

Make sure the `astsa` package is loaded and then do the following commands to look at the `salmon` dataset:

```
salmon
```

```
time(salmon)
```

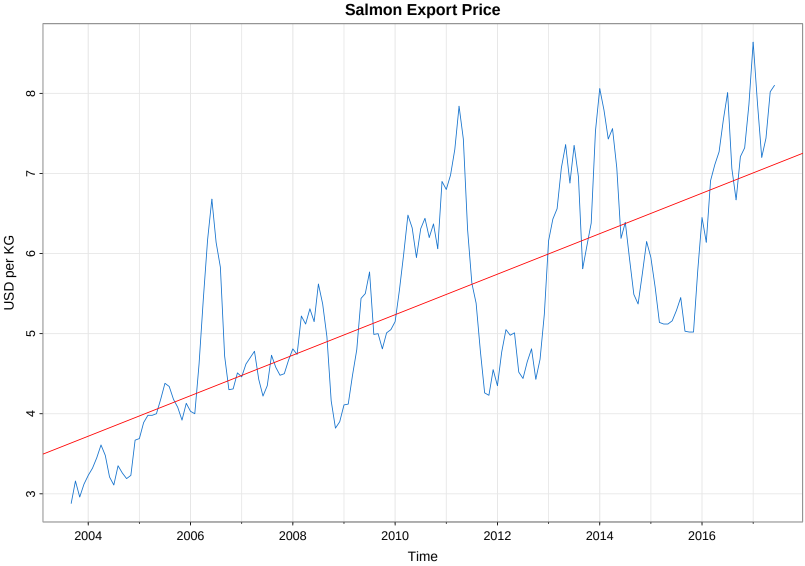
Given what you see, what are the values for x_1 and z_1?

Example: Estimating the Linear Trend of a Commodity

Now, we'll perform linear regression using the R command **lm()**.

```
##
## Call:
## lm(formula = salmon ~ time(salmon))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.69187 -0.62453 -0.07024  0.51561  2.34959
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -503.08947   34.44164  -14.61   <2e-16 ***
## time(salmon)    0.25290    0.01713   14.76   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '
##
## Residual standard error: 0.8814 on 164 degrees of freedom
## Multiple R-squared:  0.5706, Adjusted R-squared:  0.568
## F-statistic: 217.9 on 1 and 164 DF,  p-value: < 2.2e-16
```

Example: Estimating the Linear Trend of a Commodity



Example: Estimating the Linear Trend of a Commodity

What's the equation of the regression line we fit?

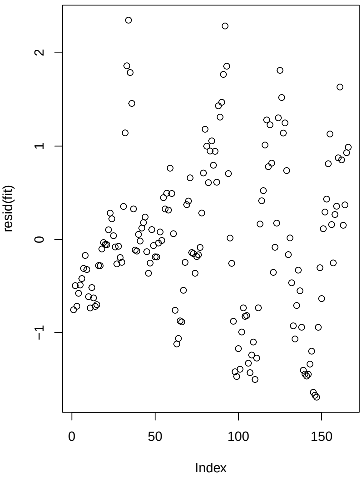
```
##
## Call:
## lm(formula = salmon ~ time(salmon))
##
## Coefficients:
##  (Intercept)  time(salmon)
##    -503.0895        0.2529
```

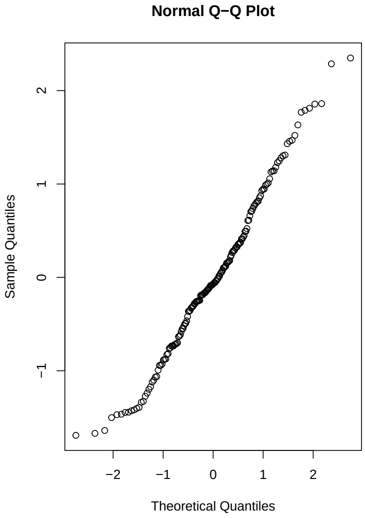
So, the ordinary least squares "best fit" regression line is:

$$x_t = -503.0895 + 0.2529 z_t$$

Example: Estimating the Linear Trend of a Commodity

How good was the assumption that the error term is modeled as white noise? Let's look at the residuals…





Back to the General Situation. . . Multiple Regression for Time Series

Suppose x_t is a time series and we're fitting the *linear regression model*

$$x_t = \beta_0 + \beta_1 z_{t1} + \beta_2 z_{t2} + \cdots + \beta_q z_{tq} + w_t$$

where $z_{t1}, z_{t2}, \ldots, z_{tq}$ are independent variables and $\{w_t\}$ is the error term, which we assume is white noise with variance σ_w^2.

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where $z_{t1}, z_{t2}, \ldots, z_{tq}$ are independent variables and $\{w_t\}$ is the error term, which we assume is white noise with variance σ_w^2.

Ordinary least squares (OLS) regression finds the regression coefficients $\beta_0, \beta_1, \ldots, \beta_q$ that minimize the error sum of squares:

$$S = \sum_{t=1}^{n} w_t^2 = \sum_{t=1}^{n} (x_t - [\beta_0 + \beta_1 z_{t1} + \beta_2 z_{t2} + \cdots + \beta_q z_{tq}])^2$$

The minimized error sum of squares is:

$$\text{SSE} = \sum_{t=1}^{n} (x_t - \hat{x_t})^2,$$

where $\hat{x_t} = \hat{\beta}_0 + \hat{\beta}_1 z_{t1} + \hat{\beta}_2 z_{t2} + \cdots + \hat{\beta}_q z_{tq}$ and $\hat{\beta}_i$ is the (OLS) regression estimate of β_i.

An unbiased estimator for the variance parameter σ_w^2 is:

$$s_w^2 = \text{MSE} = \frac{\text{SSE}}{n - (q+1)},$$

where MSE denotes the **mean square error**.

Standard Error of the Regression Coefficients

We denote the standard error for the estimated regression coefficient $\hat{\beta}_i$ as: $\mathrm{se}(\hat{\beta}_i)$. If the errors are normal, the statistic

$$t = \frac{\hat{\beta}_i - \beta_i}{\mathrm{se}(\hat{\beta}_i)}$$

will follow a t-distribution with $n - (q + 1)$ degrees of freedom.

Why do we care? Because this can be used to do hypothesis testing on whether the regression coefficient β_i is 0... which will tell us whether the corresponding independent variable z_{ti} is really needed in the model.

Hypothesis Testing to Select Best Subset of Independent Variables

Suppose a proposed model specifies that only a subset $r < q$ independent variables, say

$$\{z_{t1}, z_{t2}, \ldots, z_{tr}\}$$

is influencing the dependent variable x_t.

So we want to devise a test to see if this reduced model

$$x_t = \beta_0 + \beta_1 z_{t1} + \beta_2 z_{t2} + \cdots \beta_r z_{tr} + w_t$$

is adequate.

So, we want to do a hyptothesis test against the full model, where the null hypothesis is:

$$H_0 : \beta_{r+1} = \cdots = \beta_q = 0.$$

Hypothesis Testing to Select Best Subset of Independent Variables

The statistic we use for our hyptothesis test is the F-statistic:

$$F = \frac{(\mathrm{SSE}_r - \mathrm{SSE})/(q-r)}{\mathrm{SSE}/(n-q-1)} = \frac{\mathrm{MSR}}{\mathrm{MSE}}$$

where SSE_r is the error sum of squares under the reduced model and the other error statistics are for the full model.

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If the null hypthothesis

$$H_0 : \beta_{r+1} = \cdots = \beta_q = 0$$

is true, then $\mathrm{SSE}_r \approx \mathrm{SSE}$. Under this hyptothesis, the F-statistic has a central F-distribution with $q-r$ and $n-q-1$ degrees of freedom.

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The hull hypothesis is rejected at level α if $F > F^{q-r}_{n-q-1}(\alpha)$, the $(1-\alpha)$ percentile of the F-distribution with $q-r$ numerator and $n-q-1$ denominator degrees of freedom.

Special Case

A special case of interest is

$$H_0 = \beta_1 = \cdots = \beta_q = 0.$$

In this case the reduced model becomes

$$x_t = \beta_0 + w_t.$$

The residual sum of squares under this reduced model is:

$$\mathrm{SSE}_0 = \sum_{t=1}^{n} (x_t - \bar{x})^2$$

and SSE_0 is often called SST (*adjusted total sum of squares*).

Recall from our Regression Refresher that

$$R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}}$$

is the **coefficient of determination**. This measures the proportion of variation accounted for by all the variables (in the original model).

Stepwise Regression

- In statistical modeling, in general, we always seek to find the most *accurate* model with the least amount of *complexity*. This follows the principle of *parsimony* (or *Occam's razor*).

- For regression, this means we're looking for the model that has the best fit with the fewest number of parameters.

- We can use the hypothesis testing approach, in a systematic manner, to leave variables out and measure the impact on accuracy and fit.

Accuracy and Complexity Measures

- We use the error sum of squares

$$\mathrm{SSE} = \sum_{t=1}^{n} (x_t - \hat{x}_t)^2$$

  to measure **accuracy** of a model since it measures how close the fitted values ($\hat{x}_t$) are to the actual data (x_t).

- For a normal regression model with k coefficients, we also have the maximum likelihood estimator for the variance as:

$$\hat{\sigma}_k^2 = \frac{\mathrm{SSE}(k)}{n},$$

  where $\mathrm{SSE}(k)$ indicates the residual sum of squares under the model with k regression coefficients. The **complexity** of the model can be characterized by k, the number of parameters in the model.

Akaike's Information Criteria (AIC)

- Seeking to balance the accuracy of the fit of a model against the number of parameters in the model, we look to the notion of an **information criteria**. Here's one (due to Akaike, 1974):

Definition (Akaike's Information Criterion (AIC))

$$\mathrm{AIC} = \log \hat{\sigma}_k^2 + \frac{n+2k}{n}.$$

- We then note that a *parsimonious model* will be an accurate one (with small error $\hat{\sigma}_k^2$) that is not overly complex (small k). So we seek to minimize the AIC to specify the "best" model using this criteria.

- Note that "AIC" stands for "An Information Criterion"

Other Information Criteria

- Finding the “right” penalty term in the information criteria is a subject of some debate. So here are a few alternative criteria:

Definition (AIC, Bias Corrected(AICc))

$$\mathrm{AICc} = \log \hat{\sigma}_k^2 + \frac{n+k}{n-k-2}.$$

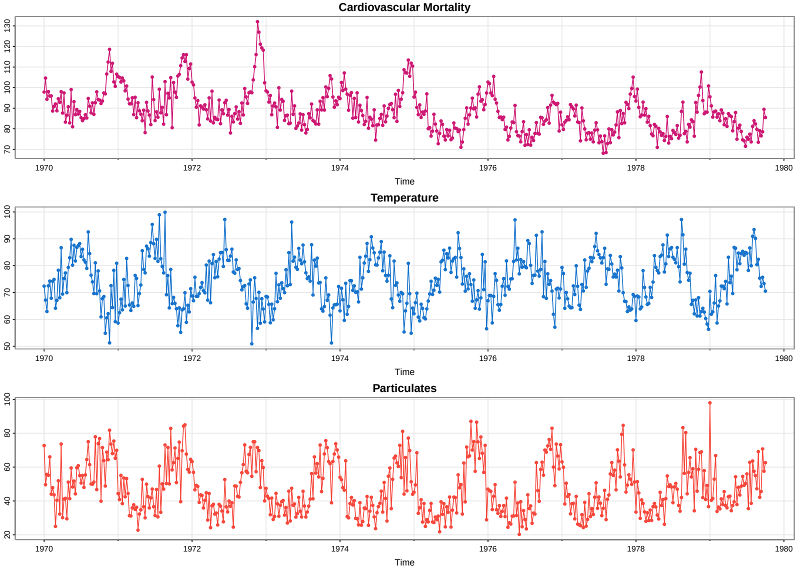
Definition (Bayesian Information Criterion, (BIC))

$$\mathrm{BIC} = \log \hat{\sigma}_k^2 + \frac{k \log n}{n}.$$

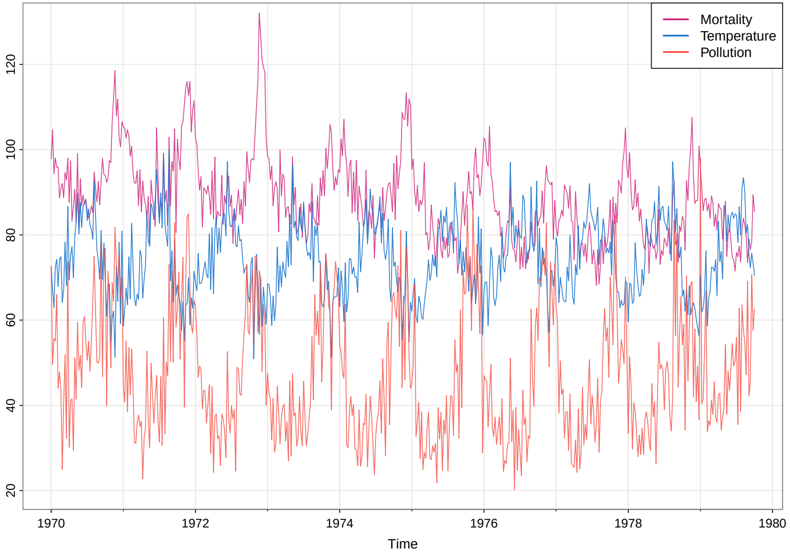
The BIC is also called the **Schwarz Information Criterion (SIC).**

- Studies indicate that the BIC does well for large samples while AICc tends to be better for small samples where the relative number of parameters is large.

Example: Los Angeles Pollution Study, 1970s



Example: Los Angeles Pollution Study, 1970s

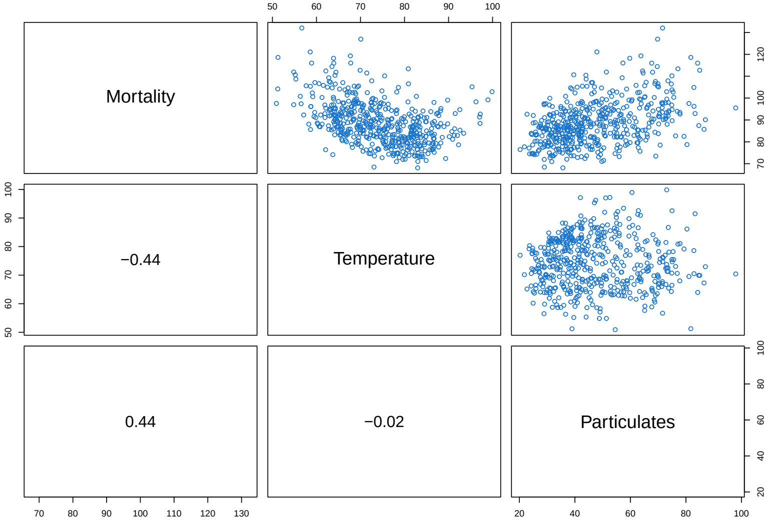


Example: Los Angeles Pollution Study, 1970s

- These time series are from a study on the possible effects of temperature and pollution on weekly mortality in Los Angeles County during the 1970s
- We notice strong seasonal components in all 3 time series
- There's a slight downward trend in the cardiovascular mortality in the 10 year period
- There appears to be an inverse relationship between mortality and temperature (mortality rate is higher for cooler temperatures; lower for warmer temps)
- There also appears to be a possible direct relationship between particulates (pollution) and mortality

Example: Los Angeles Pollution Study, 1970s

To investigate further, we look at the correlation between each of the variables in pairs, and plot their scatterplots also.



Example: Los Angeles Pollution Study, 1970s

Observations from this plot...

- We see the positive correlation between Mortality and Particulates (pollution); and this correlation looks pretty linear in the scatterplot
- We see the negative correlation between Mortality and Temperature; but this correlation does NOT look linear (it curves down and then slightly up again)
- Temperature and Particulates are nearly uncorrelated

Example: Los Angeles Pollution Study, 1970s

Introducing some notation for the 3 different time series:

Let M_t = cardiovascular mortality

Let T_t = temperature

Let P_t = particulate levels

We want to try to model M_t using a linear regression. From our analysis so far it seems clear that both T_t and P_t are influencing M_t. But we're going to look at several models for comparison.

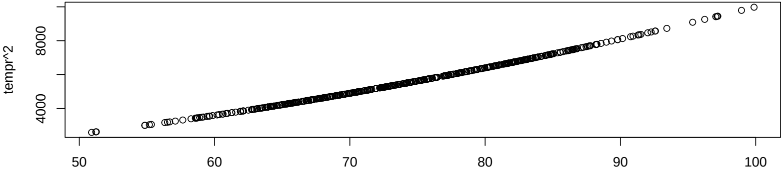
Example: Los Angeles Pollution Study, 1970s

Let's think about that non-linearity we saw in the correlation between Mortality and Temperature...

- Because of this we'd like to include a quadratic factor (T_t^2) instead of just a linear one (T_t).
- But given the temperature range we're looking at, T_t and T_t^2 are highly collinear

```r
par(mfrow=2:1)
plot(tempr, tempr^2)
cor(tempr, tempr^2)
```

```
## [1] 0.9972099
```



Example: Los Angeles Pollution Study, 1970s

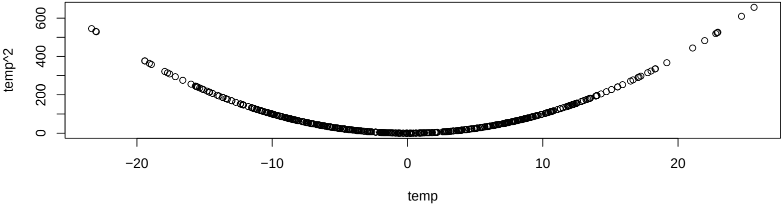
To fix this, we'll adjust for the mean temperature.

Let $T_.$ denote the mean temperature (which for this data is 74.26)

We'll look at variables $(T_t - T_.)$ and $(T_t - T_.)^2$ instead. Note they are NOT collinear in this range:

```
par(mfrow=2:1)
temp = tempr-mean(tempr)
plot(temp, temp^2)
cor(temp,temp^2)
```

```
## [1] 0.07617904
```



Example: Los Angeles Pollution Study, 1970s

So, given this, we'll test the following 4 models:

$$M_t = \beta_0 + \beta_1 t + w_t \tag{1}$$
$$M_t = \beta_0 + \beta_1 t + \beta_2 (T_t - T_\cdot) + w_t \tag{2}$$
$$M_t = \beta_0 + \beta_1 t + \beta_2 (T_t - T_\cdot) + \beta_3 (T_t - T_\cdot)^2 + w_t \tag{3}$$
$$M_t = \beta_0 + \beta_1 t + \beta_2 (T_t - T_\cdot) + \beta_3 (T_t - T_\cdot)^2 + \beta_4 P_t + w_t \tag{4}$$

(1) is a trend only model

(2) adds a linear temperature term

(3) includes the curvilinear (quadratic) temperature term

(4) adds the pollution term

Example: Los Angeles Pollution Study, 1970s

Here some R code to define the different variables...and fit the different models

```
t = time(cmort);              #  time (pure trend)
Tadj = tempr-mean(tempr);  #  (T_t - T_.)  temp adjusted by mean
Tadj2 = Tadj^2;              #  (T_t - T_.)^2
Pt = part;                   #  Particulates (pollution)
#
#
fit1 <- lm(cmort ~ t, na.action=NULL)
fit2 <- lm(cmort ~ t + Tadj, na.action=NULL)
fit3 <- lm(cmort ~ t + Tadj + Tadj2, na.action=NULL)
fit4 <- lm(cmort ~ t + Tadj + Tadj2 + Pt, na.action=NULL)


summary(fit1)
```

```
##
## Call:
## lm(formula = cmort ~ t, na.action = NULL)
##
```

Example: Los Angeles Pollution Study, 1970s

Exercise 3b.2a

Try some of the `lm()` commands and then look at the results using the `summary(fit)` and `anova(fit)`.

Note that in the `anova()` output, SSE is the "Sum Sq" of the Residuals and MSE is the "Mean Sq" of the Residuals.

Example: Los Angeles Pollution Study, 1970s

R has built-in functions to compute AIC and BIC from the output of an **lm()** command but they differ a bit from the definition of AIC and BIC our book uses. Here's some code that extracts the data and makes the needed adjustment:

```
num = length(cmort);
t = time(cmort);
fit1 <- lm(cmort ~ t, na.action=NULL)
AIC(fit3)/num - log(2*pi)
```

```
## [1] 5.028611
```

```
BIC(fit3)/num - log(2*pi)
```

```
## [1] 5.070249
```

Example: Los Angeles Pollution Study, 1970s

Exercise 3b.2b

Using this methodology, compute the AIC and BIC values for the subsequent models we fit.

Here's a table containing some of the output you should be seeing in your results:

Model	k	SSE	df	MSE	R^2	AIC	BIC
(1)	2	40,020	506	79.0	.21	5.38	5.40
(2)	3	31,413	505	62.2	.38	5.14	5.17
(3)	4	27,985	504	55.5	.45	5.03	5.07
(4)	5	20,508	503	40.8	.60	4.72	4.77

Example: Los Angeles Pollution Study, 1970s

- Examining the table of results we see that model (4) is the "best" since it has the highest R^2 with the lowest AIC/BIC values.

- Looking at the output for model (4) we see our "best fit" mortality regression model was:

$$\hat{M}_t = 2831.5 - 1.396t - .472(T_t - 74.26) + 0.023(T_t - 74.26)^2 + .255P_t$$

- Note that:
  - a negative trend is present over time
  - coefficient of adjusted temperature is also negative
  - pollution is weighted positively (can be interpreted as incremental contribution to daily deaths per unit of particulate pollution)

Example: Los Angeles Pollution Study, 1970s

- An important final note is that we should really be checking the residuals $\hat{w}_t = M_t - \hat{M}_t$, too...

- It turns out there is a fair amount of autocorrelation in these particular residuals (i.e., they are not "white"), which is an issue we'll return to later in the course.

- Try: **plot(resid(fit4), type="o")** to see a plot of the residuals for the final model.

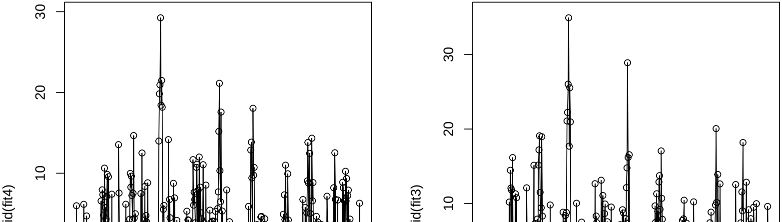
Example: Regression with Lagged Variables

Recall our correlation analysis of the SOI (Southern Oscillation Index) vs Recruitment (new fish) which indicated that the SOI at time $t-6$ months was associated/correlated with Recruitment at time t.

We'll dig deeper into this later but for now let's try fitting a naive linear regression model:

$$R_t = \beta_0 + \beta_1 S_{t-6} + w_t.$$

The R library **dynlm** fits dynamic linear models and time series regressions. For lagged regression, the set-up is faster and more intuitive.



Example: Regression with Lagged Variables

```
##
## Time series regression with "ts" data:
## Start = 1950(7), End = 1987(9)
##
## Call:
## dynlm(formula = rec ~ L(soi, 6))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -65.187 -18.234   0.354  16.580  55.790
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   65.790      1.088   60.47   <2e-16 ***
## L(soi, 6)    -44.283      2.781  -15.92   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '
##
## Residual standard error: 22.5 on 445 degrees of freedom
## Multiple R-squared:  0.3629, Adjusted R-squared:  0.3615
## F-statistic: 253.5 on 1 and 445 DF,  p-value: < 2.2e-16
```

Example: Regression with Lagged Variables

The resulting fit model is:

$$\hat{R}_t = 65.79 - 44.28 S_{t-6}$$

but taking a look at the residuals, there's a bit too much pattern to think this model is "right"... (stay tuned!)